18 There are 10 pens in a box.

3 of the pens are blue.
5 of the pens are red.
2 of the pens are green.

Alice takes at random 2 pens from the box.

Work out the probability that there are still more blue pens than green pens in the box.

Blue left > green left means 3 − b > 2 − g.

So the number of blue pens taken must be ≤ green taken.

Allowed colour pairs: RR, RG, GR, GG, BG, GB.

1 mark
Without replacement, the second denominator is 9.

P(RR) = 5/10 × 4/9 = 20/90

1 mark
Multiply along each ordered pair.

P(RG) + P(GR) = 2 × 5/10 × 2/9 = 20/90

P(GG) = 2/10 × 1/9 = 2/90

P(BG) + P(GB) = 2 × 3/10 × 2/9 = 12/90

1 mark
Add all and only the successful pairs.

Total = (20 + 20 + 2 + 12)/90 = 54/90 = 3/5

1 mark
The required probability is 0.6.

0.6

(Total for Question 18 is 4 marks)

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